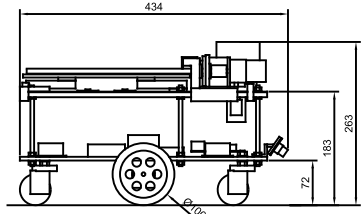
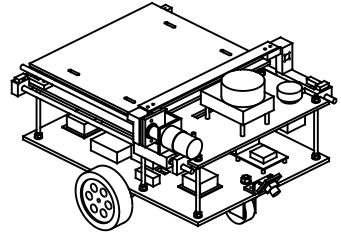


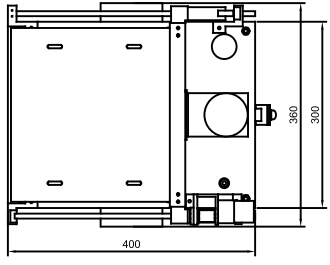
FRONT VIEW



SIDE VIEW (from the right) FRONT -->



ISOMETRIC VIEW

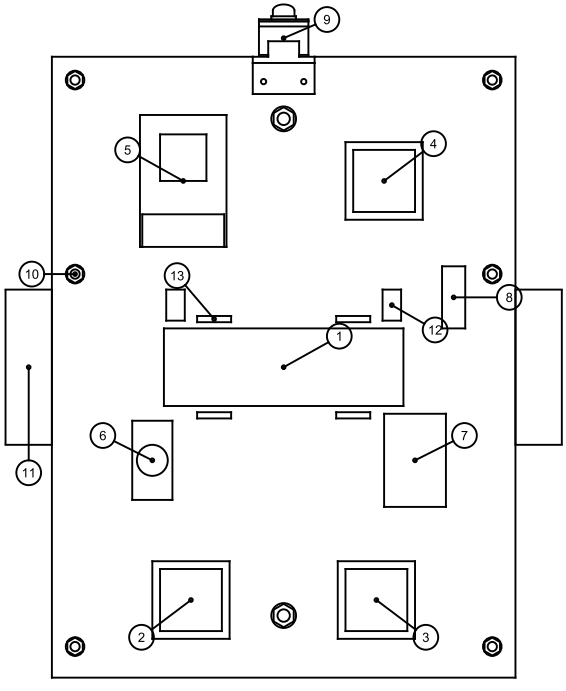


PLAN VIEW FRONT -->

NOTES

1. All dimensions in mm. Weights and loads in kg.
2. Drive: 2 x JGB37-520 12 V 170 RPM gear motors, wheels D100 x 30.
3. Casters: 2 x 50 mm swivel, M8 threaded stem (height set by nuts).
4. Frame: 6 mm plywood + 5 mm acrylic on 6 x M6 threaded rod.
5. Payload: 4 x 10 kg load cells under the cargo plate, max 15 kg.
6. Sensors: RPLIDAR A1M8 (front), USB camera tilted 35 deg down.
7. Pusher: blade on a T8 lead screw pushes the cargo off the rear (shown parked; sheet 07).
8. Sheets: 01 GA, 02 layout, 03-05 plates, 06 prints, 07 pusher, 08 BOM.

PROJECT: AUTONOMOUS CARGO ROBOT - DESIGN v3					
TITLE: GENERAL ARRANGEMENT					DWG No.: CR-V3-001
ARAB ACADEMY FOR SCIENCE, TECHNOLOGY & MARITIME TRANSPORT - GRADUATION PROJECT II					
SCALE: 1:5	SHEET: 1 / 8	UNITS: mm	PROJECTION: 1st ANGLE	DATE: 13/10/2026	DRAWN:



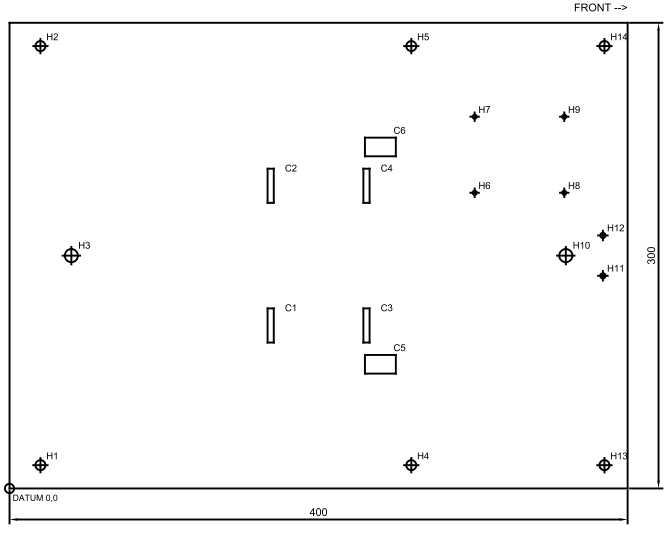
PLAN VIEW - DECK, CARGO PLATE, LIDAR AND PUSHER REMOVED (front of robot at the top)

No.	COMPONENT
1	3S LiPo battery 5000 mAh (lies across, slides out sideways)
2	BTS7960 motor driver - left motor
3	BTS7960 motor driver - right motor
4	BTS7960 motor driver - pusher motor
5	Raspberry Pi 5 on M2.5 standoffs
6	Buck converter 12 V -> 5.1 V
7	WAGO power buses (+12 V main / motor / GND)
8	Inline fuse 10 A
9	USB camera on printed mount
10	M6 threaded rod (x6)
11	Drive wheel D100 (motor under the plate)
12	Motor / encoder cable slot (x2)
13	Battery strap slots (x4)

NOTES

Small boards are held with VHB tape; the Pi on M2.5 standoffs.
Keep 10 cm slack on every wire that goes up to the deck.
Wiring: see BUILD_GUIDE.md, weeks 2 and 3.

PROJECT: AUTONOMOUS CARGO ROBOT - DESIGN v3					
TITLE: LAYOUT - DECK REMOVED					DWG No.: CR-V3-002
ARAB ACADEMY FOR SCIENCE, TECHNOLOGY & MARITIME TRANSPORT - GRADUATION PROJECT II					
SCALE: 1:2	SHEET: 2 / 8	UNITS: mm	PROJECTION: 1st ANGLE	DATE: 13/10/2026	DRAWN:



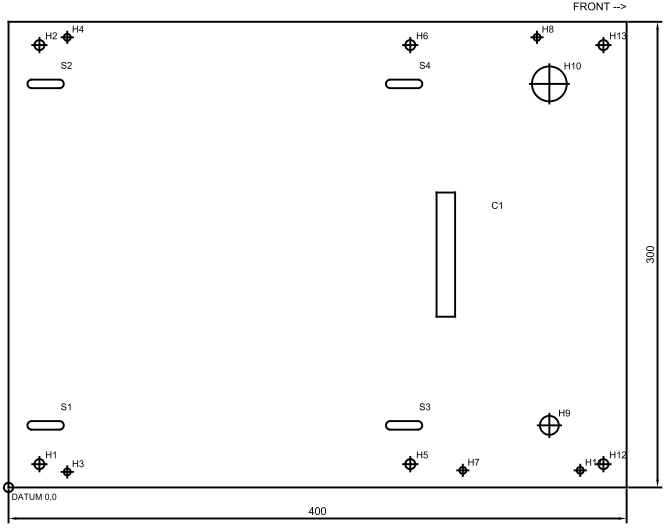
ID	X	Y	SIZE	FOR
H1	20.0	15.0	D6.5	M6 threaded rod
H2	20.0	285.0	D6.5	M6 threaded rod
H3	40.0	150.0	D8.5	caster stem M8
H4	260.0	15.0	D6.5	M6 threaded rod
H5	260.0	285.0	D6.5	M6 threaded rod
H6	301.0	190.5	D2.7	Raspberry Pi M2.5
H7	301.0	239.5	D2.7	Raspberry Pi M2.5
H8	359.0	190.5	D2.7	Raspberry Pi M2.5
H9	359.0	239.5	D2.7	Raspberry Pi M2.5
H10	360.0	150.0	D8.5	caster stem M8
H11	384.0	137.0	D3.4	camera mount M3
H12	384.0	163.0	D3.4	camera mount M3
H13	385.0	15.0	D6.5	M6 threaded rod
H14	385.0	285.0	D6.5	M6 threaded rod
C1	169.0	105.0	4 x 22	battery strap
C2	169.0	195.0	4 x 22	battery strap
C3	231.0	105.0	4 x 22	battery strap
C4	231.0	195.0	4 x 22	battery strap
C5	240.0	80.0	20 x 12	motor/encoder cables
C6	240.0	220.0	20 x 12	motor/encoder cables

X, Y = centre of the feature from DATUM 0,0 (H = hole, S = slot, C = cut-out)

NOTES

1. Material: 6 mm birch plywood (or 5 mm aluminium), 1 off.
2. Laser-cut from cad/output/dxf/bottom_plate.dxf. All dimensions in mm, +-0.2.
3. Motor-bracket holes (4 x D4.5) are drilled on assembly - see BUILD_GUIDE week 1.

PROJECT: AUTONOMOUS CARGO ROBOT - DESIGN v3					
TITLE: BOTTOM PLATE					DWG No.: CR-V3-003
ARAB ACADEMY FOR SCIENCE, TECHNOLOGY & MARITIME TRANSPORT - GRADUATION PROJECT II					
SCALE: 1:2	SHEET: 3 / 8	UNITS: mm	PROJECTION: 1st ANGLE	DATE: 13/10/2026	DRAWN:



ID	X	Y	SIZE	FOR
H1	20.0	15.0	D6.5	M6 threaded rod
H2	20.0	285.0	D6.5	M6 threaded rod
H3	38.0	10.0	D4.5	pusher bracket M4
H4	38.0	290.0	D4.5	pusher bracket M4
H5	260.0	15.0	D6.5	M6 threaded rod
H6	260.0	285.0	D6.5	M6 threaded rod
H7	294.0	11.0	D4.5	pusher bracket M4
H8	342.0	290.0	D4.5	pusher bracket M4
H9	350.0	40.0	D12.2	master switch
H10	350.0	260.0	D22.5	E-stop 22 mm
H11	370.0	11.0	D4.5	pusher bracket M4
H12	385.0	15.0	D6.5	M6 threaded rod
H13	385.0	285.0	D6.5	M6 threaded rod
S1	24.0	40.0	23.5 x 5.5 (R2.75)	load-cell screws
S2	24.0	260.0	23.5 x 5.5 (R2.75)	load-cell screws
S3	256.0	40.0	23.5 x 5.5 (R2.75)	load-cell screws
S4	256.0	260.0	23.5 x 5.5 (R2.75)	load-cell screws
C1	283.0	150.0	12 x 80	wiring pass-through

X, Y = centre of the feature from DATUM 0,0 (H = hole, S = slot, C = cut-out)

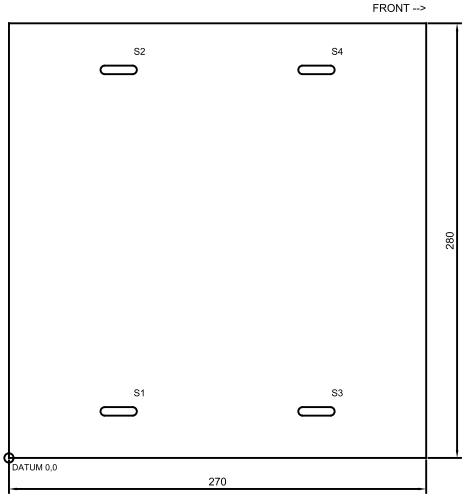
NOTES

1. Material: 5 mm clear acrylic (keep the protective film), 1 off.
2. Laser-cut from cad/output/dxf/top_deck.dxf. All dimensions in mm, +-0.2.
3. Lidar holes (4 x D2.7) are drilled on assembly, using the lidar as template.
4. The 5 x D4.5 holes are new in v3 (pusher). Deck already cut to v2? Drill them by hand.

PROJECT: AUTONOMOUS CARGO ROBOT - DESIGN v3					
TITLE: TOP DECK					DWG No.: CR-V3-004
ARAB ACADEMY FOR SCIENCE, TECHNOLOGY & MARITIME TRANSPORT - GRADUATION PROJECT II					
SCALE: 1:2	SHEET: 4 / 8	UNITS: mm	PROJECTION: 1st ANGLE	DATE: 13/10/2026	DRAWN:

ID	X	Y	SIZE	FOR
S1	71.0	30.0	23.5 x 5.5 (R2.75)	load-cell screws
S2	71.0	250.0	23.5 x 5.5 (R2.75)	load-cell screws
S3	199.0	30.0	23.5 x 5.5 (R2.75)	load-cell screws
S4	199.0	250.0	23.5 x 5.5 (R2.75)	load-cell screws

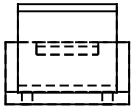
X, Y = centre of the feature from DATUM 0,0 (H = hole, S = slot, C = cut-out)



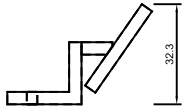
NOTES

1. Material: 5 mm clear acrylic, 1 off. Leave it bare: the cargo must slide (no mat).
2. Laser-cut from cad/output/dxf/cargo_plate.dxf. All dimensions in mm, +-0.2.
3. Rests only on the 4 load cells (slots S1-S4). Same plate as v2.

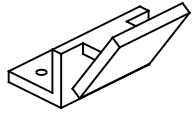
PROJECT: AUTONOMOUS CARGO ROBOT - DESIGN v3					
TITLE: CARGO PLATE					DWG No.: CR-V3-005
ARAB ACADEMY FOR SCIENCE, TECHNOLOGY & MARITIME TRANSPORT - GRADUATION PROJECT II					
SCALE: 1:2	SHEET: 5 / 8	UNITS: mm	PROJECTION: 1st ANGLE	DATE: 13/10/2026	DRAWN:

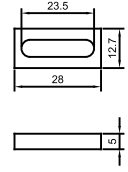


FRONT

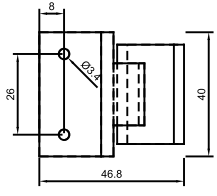


CAMERA MOUNT - SIDE





LOAD-CELL SPACER (8 off), slot 5.5 wide



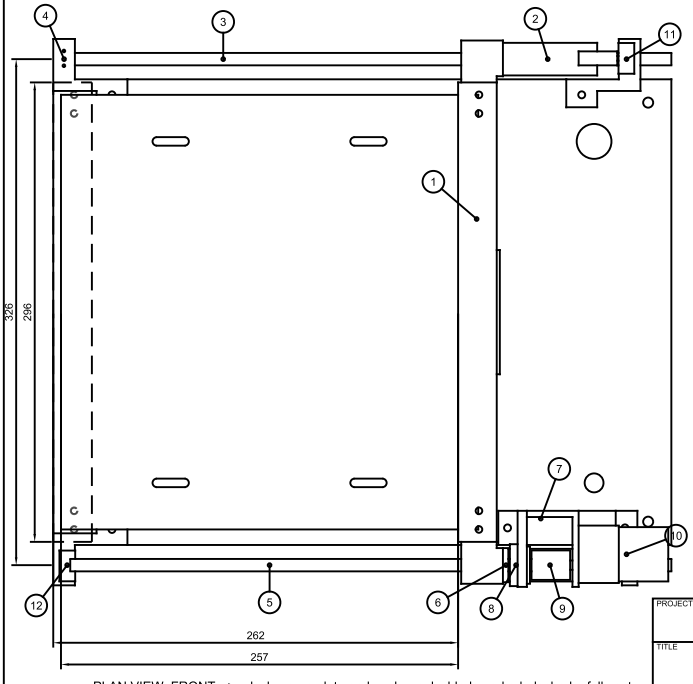
PLAN

Holder face tilted 35 deg down. Drill 2 mm holes to suit the camera board.

NOTES

1. Material PETG, 0.2 mm layers, 4 walls.
2. Camera mount: 1 off, 40 % infill, supports on build plate only.
3. Load-cell spacer: 8 off, 100 % infill, print flat.
4. Pusher prints (2 carriages, 3 rod holders, 1 drive bracket): sheet 07.
5. STL files: cad/output/parts/

PROJECT: AUTONOMOUS CARGO ROBOT - DESIGN v3					
TITLE: PRINTED PARTS				DWG No.: CR-V3-006	
ARAB ACADEMY FOR SCIENCE, TECHNOLOGY & MARITIME TRANSPORT - GRADUATION PROJECT II					
SCALE: 1:1	SHEET: 6 / 8	UNITS: mm	PROJECTION: 1st ANGLE	DATE: 13/10/2026	DRAWN:

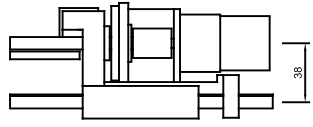


PLAN VIEW FRONT --> deck, cargo plate and pusher only; blade parked, dashed = fully out

No.	COMPONENT	QTY
1	Push blade: aluminium angle 25x25x2, 296 long	1
2	Carriage, printed (L with the switch finger, R with the nut), 2 x LM8UU each	2
3	Guide rod: smooth steel D8 x 400	2
4	Rod holder, printed (A: rear right + front left, B: rear left)	3
5	Lead screw T8, lead 8, cut to 310	1
6	Brass T8 flange nut (in the right carriage)	1
7	Drive bracket, printed	1
8	KFL08 flange bearing block (takes the push force)	1
9	Flexible coupler 6 / 8 mm	1
10	Gear motor JGB37-520 12 V, about 110 RPM	1
11	Limit switch HOME, on the front-left rod holder (pressed when parked)	1
12	Limit switch END, on the rear-right rod holder (pressed when fully out)	1

NOTES

1. Stroke 262 mm; the blade ends flush with the rear edge.
2. The blade clears the cargo plate by 3 mm: it must never touch it (the plate rests on the load cells).
3. Everything stays below 245 mm, under the lidar's scan plane.
4. Holders and bracket clip over the deck edge; 5 x M4 through the deck.
5. Leave the cargo plate bare (no anti-slip mat) so the cargo slides.
6. Conveyor: belt 180-205 mm above the floor, with a backstop taller than 260 mm so the lidar can see it.
7. Printed parts: PETG, 4 walls, 40 % infill. STL: cad/output/parts/



DETAIL - DRIVE END (seen from the right) FRONT -->

PROJECT: AUTONOMOUS CARGO ROBOT - DESIGN v3

TITLE: PUSHER ASSEMBLY

DWG No.: CR-V3-007

ARAB ACADEMY FOR SCIENCE, TECHNOLOGY & MARITIME TRANSPORT - GRADUATION PROJECT II

SCALE	SHEET	UNITS	PROJECTION	DATE	DRAWN
1:2	7 / 8	mm	1st ANGLE	13/10/2026	

No.	GROUP	DESCRIPTION	QTY	MAKE / BUY
1	Compute	Raspberry Pi 5 8GB + Active Cooler + 64GB A2 microSD (x2)	1	buy
2	Drive	JGB37-520 12V 170 RPM gear motor with Hall encoder (6 wires)	2	buy
3	Drive	Steel mounting bracket for 37 mm gear motor (+ its M3 screws)	2	buy
4	Drive	Wheel 100 mm x 30 mm with 6 mm D-shaft hub	2	buy
5	Drive	Swivel caster 50 mm, M8 threaded stem, >= 20 kg rating	2	buy
6	Drive	BTS7960 (IBT-2) 43 A motor driver	2	buy
7	Sensors	RPLIDAR A1M8 (with USB adapter)	1	buy
8	Sensors	USB (UVC) camera board 32x32 mm, wide-angle M12 lens	1	buy
9	Sensors	Camera mount (camera_mount.stl)	1	3D print PETG
10	Payload	Straight-bar load cell 10 kg, 80x12.7x12.7 mm	4	buy
11	Payload	HX711 module	4	buy
12	Payload	Load-cell spacer (load_cell_spacer.stl)	8	3D print PETG 100 %
13	Payload	Cargo plate 270x280x5 (cargo_plate.dxf), left bare so the cargo can slide	1	laser-cut acrylic
14	Pusher	JGB37-520 12 V gear motor, about 110 RPM, 6 mm D-shaft (no encoder needed)	1	buy
15	Pusher	BTS7960 (IBT-2) 43 A motor driver	1	buy
16	Pusher	T8 lead screw, 8 mm lead, 350 mm, with brass flange nut - cut to 310 mm	1	buy
17	Pusher	Smooth steel rod 8 mm x 400 mm (linear shaft), not cut	2	buy
18	Pusher	LM8UU linear ball bearing	4	buy
19	Pusher	KFL08 flange bearing block, 8 mm bore	1	buy
20	Pusher	Flexible shaft coupler 6 mm to 8 mm (D19 x L25)	1	buy
21	Pusher	Lever microswitch, subminiature 20x10x6.4 mm (KW11 / SS-5GL type)	2	buy
22	Pusher	Aluminium angle 25x25x2 mm, cut to 296 mm (pusher_blade.step)	1	cut from 1 m bar
23	Pusher	Carriage (pusher_carriage_L.stl, pusher_carriage_R.stl)	2	3D print PETG
24	Pusher	Rod holder (pusher_rail_holder_A.stl x 2, pusher_rail_holder_B.stl x 1)	3	3D print PETG
25	Pusher	Drive bracket (pusher_drive_bracket.stl)	1	3D print PETG
26	Power	3S LiPo 11.1 V 5000 mAh XT60 + balance charger + LiPo bag + voltage alarm	1	buy
27	Power	XL4015 5 A buck converter + USB-C pigtail	1	buy
28	Power	Inline fuse holder + 10 A fuses, WAGO 221 connectors, XT60 pairs	1	buy
29	Power	Emergency stop 22 mm panel-mount, latching, NC contact block	1	buy
30	Power	Toggle switch 12 mm panel-mount, 12 V DC >= 15 A	1	buy
31	Chassis	Bottom plate 400x300x6 (bottom_plate.dxf)	1	laser-cut 6 mm plywood
32	Chassis	Top deck 400x300x5 (top_deck.dxf)	1	laser-cut 5 mm acrylic
33	Chassis	M6 threaded rod, cut to 130 mm	6	cut from 1 m rod
34	Chassis	M6 nut + M6 washer	24	buy
35	Chassis	M8 nut + M8 washer (casters)	4	buy
36	Hardware	M4x12 screws + nylock nuts (motor brackets), M4/M5x16 (load cells), M3/M2.5 screws + standoffs (Pi, camera), VHB tape, velcro strap, zip ties	1	buy
37	Hardware	Pusher: M4x20 + nut + washer x 10 (rod holders, blade, drive bracket, bearing block), M4x25 + nut + washer x 1 (drive bracket, under the motor), M3x16 x 4 (T8 nut), M3x8 socket-head x 4 (motor), M3x8 x 4 (rod pinch screws), M2x10 x 4 (limit switches)	1	buy

PROJECT	AUTONOMOUS CARGO ROBOT - DESIGN v3					
TITLE	BILL OF MATERIALS				DWG No.	CR-V3-008
ARAB ACADEMY FOR SCIENCE, TECHNOLOGY & MARITIME TRANSPORT - GRADUATION PROJECT II						
SCALE	SHEET	UNITS	PROJECTION	DATE	DRAWN	
-	8 / 8	mm	1st ANGLE	13/10/2026		